Derivación

$$\mathcal{L}: V_a(R_f l_f + R_I l_I) - \sum_{S=F,I} \frac{\Psi_S \, l_S^{1+1/\phi_S}}{1+\frac{1}{\phi_S}} + \lambda_H(\bar{H} - l_F - l_I)$$

CPO

$$\{l_S\} \quad V_a R_S - \frac{\Psi_S}{1+\frac{1}{\phi_S}}\left(1+\frac{1}{\phi_S}\right) l_S^{1/\phi_S} - \lambda_H = 0$$

$$\hookrightarrow \quad V_a R_S = \Psi_S \, l_S^{1/\phi_S} + \lambda_H$$

despejo l_S:

$$l_S = \frac{(V_a R_S - \lambda_H)^{\phi_S}}{\Psi_S^{\phi_S}} \qquad (6)$$

$$\frac{l_I}{l_H} = \frac{(V_a R_I - \lambda_H)^{\phi_I}}{(V_a R_F - \lambda_H)^{\phi_F}} \cdot \frac{\Psi_F^{\phi_F}}{\Psi_I^{\phi_I}}$$

Ahora del KKT

$$\lambda_H \cdot (\bar{H} - l_f - l_I) = 0$$

$\bar{H} - l_f - l_I > 0$ no binding $\rightarrow \quad \lambda_H = 0$

$$\therefore \quad l_S = \left(\frac{V_a R_S}{\Psi_S}\right)^{\phi_S}$$

$$\frac{l_I}{l_F} = \frac{(V_a R_I - \lambda_H)^{\phi_I}}{(V_a R_F - \lambda_H)^{\phi_F}} \cdot V_a^{\phi_I - \phi_F} = h \cdot V_a^{\phi_I - \phi_F}$$

Si $\phi_I = \phi_F \rightarrow \quad \frac{l_I}{l_F} = h$

Si $\phi_I \neq \phi_F \qquad \frac{l_I}{l_F} = h \cdot V_a^{\phi_I - \phi_F}$ ahora depende de V_a (depende ligeramente)